

L-functions and periods of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$
 Comments: (1) Working draft, currently restricted to $\dim S_k \leq 1$,
 (2) the properly meromorphic MF case at very preliminary stage

May 21, 2026

Abstract

For any $f \in S_{12}^!$ on $\mathrm{SL}_2(\mathbb{Z})$ (the weakly holomorphic forms of weight 12 with vanishing constant Fourier coefficient; this includes the unique normalised holomorphic cusp form $\Delta \in S_{12} \subset S_{12}^!$) we exhibit an explicit linear functional $f \mapsto (\omega^+(f), \omega^-(f)) \in \mathbb{C}^2$ given by integration of f along two finite line segments in the upper half-plane against universal $\mathbb{Q}$ -polynomial kernels. Specialising to Δ recovers Eichler–Shimura’s classical periods $\omega^\pm(\Delta)$; specialising to the unique $\hat{\Delta} \in S_{12}^!$ with $\hat{\Delta}(\tau) = q^{-1} + O(q^2)$ produces Brown’s quasi-periods $\eta^\pm(\hat{\Delta})$. The construction is the explicit Bernoulli-operator unpacking of the Diamantis–Rolen finite-endpoint cocycle; the functional is basepoint-independent. Replacing the $\mathbb{Q}$ -polynomial T -kernel by a Hurwitz-zeta function of τ promotes the integer- s formula to a finite-contour expression for the completed L -function $\Lambda^*(f, s)$ at all complex s , which agrees established L -function at all values in the critical strip. The finite-contour formulation extends without modification to meromorphic modular forms with interior poles, with a wrinkle: τ_0 -invariance is broken by discrete wall-crossing jumps that encode the residues of f at its interior poles, suggesting the natural starting point for an L -function theory of generic meromorphic modular forms.

1 Introduction and main result

The period polynomial of a weight- k holomorphic cusp form $f \in S_k$ on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ is the polynomial

$$r_f(X) := \int_0^{i\infty} f(\tau) (\tau - X)^{k-2} d\tau \in V_{k-2} \otimes \mathbb{C},$$

which decomposes into even and odd parts $r_f^\pm$ whose coefficients are $\mathbb{Q}$ -rationals multiples of $\omega^\pm(f) (2\pi i)^\bullet$, with the periods $\omega^\pm(f)$ in turn expressible as $\mathbb{Q}$ -multiples of critical L -values $L(f, s)$. The Eichler–Shimura picture realises $\omega^\pm(f)$ as $\mathbb{C}$ -linear functionals of f via a contour integral, and the resulting coordinates carry deep arithmetic.

For weakly holomorphic $f \in S_k^!$ with a cusp pole, the integral $\int_0^{i\infty}$ diverges and the classical construction fails. Nevertheless Brown [1] defines and numerically computes *quasi-periods* $\eta^\pm(f)$ for the unique form $\hat{\Delta} \in S_{12}^!$ satisfying $\hat{\Delta}(\tau) = q^{-1} + O(q^2)$.

To implement Brown’s construction, we use the Diamantis–Rolen finite-endpoint cocycle [2]: replacing the divergent contour $0 \rightarrow i\infty$ with a finite segment $\gamma^{-1}\tau_0 \rightarrow \tau_0$ in $\mathbb{H}$ produces a genuine 1-cocycle C^f of Γ whose cohomology class encodes $\omega^\pm(f)$ for holomorphic f and $\eta^\pm(f)$ for weakly holomorphic f , on the same footing. Brown’s $\omega^\pm$ and $\eta^\pm$ thus come from a single procedure. The present note unpacks this procedure to the level of an explicit linear functional, fully transparent at the coefficient level.

Conventions. Fix $k = 12$, $n = k - 2 = 10$. Let $V_n = \bigoplus_{i+j=n} \mathbb{Q}X^iY^j$ with the right Γ -action $(X, Y)|_\gamma = (aX + bY, cX + dY)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Write $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. We write $[P]_{X^{10-j}Y^j}$ for the coefficient of $X^{10-j}Y^j$ in $P \in V_{10}$, and set $q = e^{2\pi i\tau}$. We use the convention that $S_k^!$ denotes the weakly holomorphic forms of weight k with vanishing constant Fourier coefficient, so $S_k \subset S_k^!$.

Zagier's space $W_{12} \subset V_{10}$ of admissible period polynomials at weight 12 is 3-dimensional, with the standard basis

$$\begin{aligned} P_0(X, Y) &= X^{10} - Y^{10}, \\ P_1(X, Y) &= X^8Y^2 - 3X^6Y^4 + 3X^4Y^6 - X^2Y^8, \\ P_2(X, Y) &= 4X^9Y - 25X^7Y^3 + 42X^5Y^5 - 25X^3Y^7 + 4XY^9, \end{aligned} \tag{1}$$

splitting as $W_{12} = W_{12}^+ \oplus W_{12}^-$ with $W_{12}^+ = \langle P_0, P_1 \rangle$ (the 2-dimensional even-in- X sector, containing the Eisenstein direction P_0) and $W_{12}^- = \langle P_2 \rangle$ (the 1-dimensional odd-in- X cuspidal sector) [5]. Brown's basis at S [1, §8.2] is $P_S^+ = -\frac{36}{691}P_0 + P_1$ and $P_S^- = P_2$.

The two main results. First, for each interior monomial $j \in \{1, 2, \dots, 9\}$ let ε_j denote the parity sign — if j is odd, + if j is even — and let $c_S^j := [P_S^{\varepsilon_j}]_{X^{10-j}Y^j} \in \mathbb{Q}$ denote the coefficient of $X^{10-j}Y^j$ in Brown's basis vector of the appropriate parity.

Second, we extend this into an L-function to hold for a generic meromorphic modular, by generalizing off of $s \in \{1, \dots, k-1\}$ to a function in the entire s -plane in a natural way. We prove that these results exactly reproduce those in [3]. We note that the construction also includes the case of L-functions associated with meromorphic modular forms that are regular at cusps, introduced in [4]. Further, we emphasize that these newer L-function derivations naturally follow from first principles and do not depend on any regularization at all.

Theorem 1.1 (Periods and quasi-periods as a linear functional). *Focus on k such that $\dim(S_k) \leq 1$. Here, there exist universal polynomials $k_S^j, k_T^j \in \mathbb{Q}[\tau]$ of degree $\leq k-2$ in τ , indexed by $j \in \{1, \dots, k-2\}$ and, for the case of $k = 12$ are given explicitly in Table 1 below, with the following property. For any $\tau_0 \in \mathbb{H}$ and any $f \in S_k^!$,*

$$\omega^{\varepsilon_j}(f) = \frac{(2\pi i)^{k-1}}{c_S^j(-1)^j} \binom{10}{j} \left[\int_{-1/\tau_0}^{\tau_0} f(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} f(\tau) k_T^j(\tau) d\tau \right]. \tag{2}$$

The value $\omega^{\varepsilon_j}(f)$ is independent of the basepoint τ_0 and of the choice of interior monomial j (within its parity sector). Both contour integrals are finite for every $f \in S_k^!$, with no regularisation, analytic continuation, or limit required.

Theorem 1.2 (Complex- s finite-contour L-function for Δ). *For all $s \in \mathbb{C} \setminus \{1, k-1\}$ and all $\tau_0 \in \mathbb{H}$, and for every $f \in S_k^!$ where $\dim(S_k) \leq 1$, define*

$$L_f^*(s) = i^{-s} \left[\int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau + \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau \right], \tag{3}$$

where the Hurwitz-zeta T -kernel is

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1). \tag{4}$$

This $L_f^(s)$ is the L-function for the classical case of $f \in S_k$ and equals the L-functions previously defined in [3] and [4]. However, it does so without any need to regularize divergences. Further, $L_f^*(s)$ satisfies the functional equation $L_f^*(s) = i^k L_f^*(k-s)$.*

Theorem 1.1 is the analog of the classical $\omega^\pm \propto L(f, s) (2\pi i)^\bullet$ formula in finite-contour DR coordinates, valid uniformly for holomorphic and weakly holomorphic f . We prove it in §§2–6: existence of the kernels via the Bernoulli operator (§4), basepoint-independence via an explicit coboundary identity (§6), j -independence via the lockstep structure of W_{12} . Two specialisations of immediate interest:

Corollary 1.3 (Classical periods $\omega^\pm(\Delta)$). *For $f = \Delta \in S_{12} \subset S_{12}^!$, the unique normalised holomorphic cusp form, (2) reproduces the Eichler–Shimura periods. At $\tau_0 = 0.3 + 1.2i$ all nine extractions agree (relative residual $\sim 10^{-24}$) on*

$$\omega^+(\Delta) = -68\,916\,772.80959519475431\dots, \quad \omega^-(\Delta) = -5\,585\,015.379310401866877\dots i,$$

matching Brown [1, §8.3] to all 20 reported digits.

Corollary 1.4 (Quasi-periods $\eta^\pm(\hat{\Delta})$). *For $f = \hat{\Delta} = \Delta \cdot (J^2 + 24J - 393444) \in S_{12}^!$, (2) reproduces Brown’s quasi-periods. At $\tau_0 = 0.3 + 1.2i$, all nine extractions agree (relative residual $\sim 10^{-17}$) on*

$$\eta^+(\hat{\Delta}) = 127\,202\,100\,647.1770947773\dots, \quad \eta^-(\hat{\Delta}) = 10\,276\,732\,343.6491327508\dots i,$$

matching Brown [1, §8.3] to all 20 reported digits.

The integrand and contour structure are *identical* in Corollaries 1.3 and 1.4; only the modular form f under the integral sign changes. This is the precise sense in which the DR construction places $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ on the same footing.

Corollary 1.5 (Uniform applicability across $\mathbb{C}[J] \cdot \Delta$). *For any $R \in \mathbb{C}[J]$ with $a_{R\Delta}(0) = 0$, formula (2) computes $\omega^\pm(R \cdot \Delta)$ via the same universal kernels. In particular, the entire $\mathbb{C}$ -span of products $R(J) \Delta$ (with vanishing constant term) maps linearly into $H_{\text{cusp}}^1(\Gamma; V_{10} \otimes \mathbb{C}) = \mathbb{C}[P^+] \oplus \mathbb{C}[P^-]$, landing on the pair $(\omega^+, \omega^-) \in \mathbb{C}^2$ via (2).*

Why this matters. A “period” is, in classical usage, the integral of an algebraic differential along a cycle. Eichler–Shimura realises $\omega^\pm(\Delta)$ this way: as Mellin transforms of Δ on $[0, i\infty]$, packaged into a polynomial in V_{10} . Theorem 1.1 preserves the period-as-integral picture but on a finite contour in $\mathbb{H}$, with universal polynomial kernel weights k_S^j, k_T^j in place of the bare τ^j moments. For holomorphic Δ , taking $\tau_0 \rightarrow i\infty$ collapses the T -segment, sends the S -segment to $[0, i\infty]$, and reduces the kernels to bare monomials — recovering classical Mellin (Lemma 5.3). For weakly holomorphic $\hat{\Delta}$, the limit is unavailable, but at every finite $\tau_0 \in \mathbb{H}$ the formula and its value are well-defined. “Quasi-periods” are honestly periods in the same geometric sense as the classical ones: integrals of a modular form along explicit cycles in $\mathbb{H}$, against explicit polynomial kernels, with the contour bounded away from the cusp. The basepoint τ_0 — not analytic continuation, not a non-holomorphic regulator — is what tames the cusp pole.

Outline. §2 introduces the Diamantis–Rolen cocycle and its moment expansion; §3 introduces the coboundary modification by a polynomial P satisfying $P|_T - P = C_T^f$ (existence, uniqueness, Bernoulli closed form); §4 establishes the universal kernels of Theorem 1.1 and presents Table 1; §5 verifies the formula numerically for Δ and $\hat{\Delta}$ and shows it produces an exponentially convergent q -series for $\omega^\pm$; §6 interprets $\omega^\pm$ as H_{cusp}^1 coordinates and proves the basepoint and j -independence; §7 contrasts with BFK’s regularisation; §9 sketches the extension to fully meromorphic forms with interior poles and offers a gauge-theoretic analogy as a thread for future work.

2 The Diamantis–Rolen cocycle and its moment expansion

Definition 2.1 (Diamantis–Rolen finite-endpoint cocycle). For $f \in S_{12}^!$ and $\tau_0 \in \mathbb{H}$, define $C^f: \Gamma \rightarrow V_{10} \otimes \mathbb{C}$ by

$$C_\gamma^f(X, Y) := (2\pi i)^{11} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^{10} d\tau, \quad (5)$$

the contour being the straight line segment from $\gamma^{-1}\tau_0$ to τ_0 in $\mathbb{H}$.

Lemma 2.2. C^f is a 1-cocycle of Γ : $C_{\gamma_1\gamma_2}^f = C_{\gamma_1}^f|_{\gamma_2} + C_{\gamma_2}^f$ for all $\gamma_1, \gamma_2 \in \Gamma$.

Proof. Path-additivity: $\int_{(\gamma_1\gamma_2)^{-1}\tau_0}^{\tau_0} = \int_{(\gamma_1\gamma_2)^{-1}\tau_0}^{\gamma_2^{-1}\tau_0} + \int_{\gamma_2^{-1}\tau_0}^{\tau_0}$. The substitution $\tau = \gamma_2\sigma$ in the first integral, combined with weight-12 modularity of $f(\tau) d\tau$ (transforming with weight 10 under $\sigma \mapsto \gamma_2\sigma$, exactly cancelling the weight-10 slash on $(X - \tau Y)^{10}$), identifies it with $C_{\gamma_1}^f|_{\gamma_2}(X, Y)/(2\pi i)^{11}$. $\square$

A 1-cocycle of Γ is determined by its values at the generators S, T . Define the *moment integrals*

$$N_j(f, \tau_0) := \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^j d\tau, \quad M_j(f, \tau_0) := \int_{\tau_0-1}^{\tau_0} f(\tau) \tau^j d\tau, \quad j = 0, 1, \dots, 10. \quad (6)$$

Expanding $(X - \tau Y)^{10}$ binomially in (5) gives

$$C_S^f(X, Y) = (2\pi i)^{11} \sum_{j=0}^{10} (-1)^j \binom{10}{j} N_j X^{10-j} Y^j, \quad (7)$$

$$C_T^f(X, Y) = (2\pi i)^{11} \sum_{j=0}^{10} (-1)^j \binom{10}{j} M_j X^{10-j} Y^j. \quad (8)$$

For $f \in S_{12}^!$ the q -coefficients $a_f(n)$ grow at most as $e^{4\pi\sqrt{n}}$ [1], while $|q|^n = e^{-2\pi n \operatorname{Im}(\tau)}$ decays super-exponentially in n for any $\tau \in \mathbb{H}$. The Fourier series therefore converges absolutely on every compact subset of $\mathbb{H}$, and the moment integrals (6) are unambiguously finite for any $\tau_0 \in \mathbb{H}$.

Remark 2.3. The cuspidality input $a_f(0) = 0$ (built into the definition of $S_{12}^!$) enters in exactly one place: it makes $M_0(f, \tau_0) = \int_{\tau_0-1}^{\tau_0} f d\tau = a_f(0)$ (by periodicity of f), so vanishing of $a_f(0)$ is equivalent to vanishing of the X^{10} coefficient of C_T^f . This is the solvability condition for the coboundary equation in §3.

3 Cuspidality and the coboundary polynomial

The cocycle C^f encodes more data than $\omega^\pm(f)$. To extract the periods we replace C^f by a cohomologous *cuspidal* cocycle whose value at T vanishes. The price is a coboundary modification by $P \in V_{10} \otimes \mathbb{C}$ satisfying

$$P|_T - P = C_T^f. \quad (\star)$$

The modified cocycle

$$C_\gamma'^f := C_\gamma^f - (P|_\gamma - P) \quad (9)$$

is cohomologous to C^f and satisfies $C_T'^f = 0$.

Proposition 3.1 (Existence and uniqueness of P). *The operator $\delta_T: V_{10} \rightarrow V_{10}$, $P \mapsto P|_T - P$, has $\ker(\delta_T) = \mathbb{C} \cdot Y^{10}$ and $\text{im}(\delta_T) = \{Q \in V_{10} : [Q]_{X^{10}} = 0\}$. Hence $(\star)$ is solvable iff $[C_T^f]_{X^{10}} = 0$, which by Remark 2.3 is equivalent to $a_f(0) = 0$ (automatic for $f \in S_{12}^!$). The solution P is unique modulo $\mathbb{C} \cdot Y^{10}$.*

Proof. T acts on V_{10} as $X \mapsto X + Y$, $Y \mapsto Y$, fixing only powers of Y , hence $\ker(\delta_T) = \mathbb{C} \cdot Y^{10}$. Writing $P = \sum_{j=0}^{10} p_j X^{10-j} Y^j$ and expanding $(X + Y)^{10-j}$ binomially, the $X^{10-m} Y^m$ coefficient of $P|_T - P$ is $\sum_{j=0}^{m-1} \binom{10-j}{m-j} p_j$. This is upper-triangular in m : the $m = 0$ row is identically 0 (requiring $[C_T^f]_{X^{10}} = 0$), and rows $m = 1, \dots, 10$ determine $p_0, \dots, p_9$ in turn, with p_{10} unconstrained. $\square$

We fix the convention $p_{10} = 0$; alternative choices shift $C_S'^f$ by $c(X^{10} - Y^{10})$ (since $Y^{10}|_S - Y^{10} = X^{10} - Y^{10}$), a coboundary in V_{10} leaving the cohomology class and the periods invariant.

Proposition 3.2 (Bernoulli closed form for P). *With $T = e^{Y\partial_X}$ acting on V_{10} as a translation operator, the unique solution of $(\star)$ with $p_{10} = 0$ is*

$$P(X, Y) = \frac{1}{Y} \int_0^X C_T^f(t, Y) dt + \sum_{k=1}^{10} \frac{B_k}{k!} Y^{k-1} \frac{\partial^{k-1} C_T^f}{\partial X^{k-1}}(X, Y), \quad (10)$$

where B_k are Bernoulli numbers ($B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $\dots$; $B_{2\ell+1} = 0$ for $\ell \geq 1$). The series terminates at $k = 10$, with at most six nonzero Bernoulli terms.

Proof. Formally $\delta_T = e^{Y\partial_X} - 1$ as an operator, so $\delta_T^{-1} = \sum_{k \geq 0} B_k (Y\partial_X)^{k-1} / k!$ by the Bernoulli generating identity $z/(e^z - 1) = \sum_k B_k z^k / k!$. The $k = 0$ summand is $(Y\partial_X)^{-1} = \partial_X^{-1} / Y$ (the antiderivative in X divided by Y), giving the integral term; the $k \geq 1$ summands give the differential terms. On a polynomial of finite X -degree the operator series terminates. Agreement of (10) with the upper-triangular recursion in the proof of Proposition 3.1 is verified to relative residual $\sim 10^{-42}$ at 45-digit working precision in [10]. $\square$

Substituting (8) into (10) expresses each p_l as an explicit $\mathbb{Q}$ -linear combination of the moments $M_1, \dots, M_{10}$:

$$p_0 = -(2\pi i)^{11} M_1, \quad p_1 = 5(2\pi i)^{11} (M_1 + M_2), \quad p_2 = -(2\pi i)^{11} \left(\frac{15}{2} M_1 + \frac{45}{2} M_2 + 15 M_3 \right), \quad (11)$$

with $p_3, \dots, p_9$ following the same pattern.

Definition 3.3 (Modified cocycle value $C_S'^f$). *With P as in Proposition 3.2 and using $P|_S(X, Y) = P(Y, -X)$:*

$$C_S'^f(X, Y) := C_S^f(X, Y) - [P(Y, -X) - P(X, Y)].$$

This is a polynomial in $V_{10} \otimes \mathbb{C}$ whose coefficients are explicit linear combinations of the moments $\{M_j, N_j\}_{j=0}^{10}$ of f , with rational/Bernoulli weights and overall factor $(2\pi i)^{11}$.

By construction $C_T'^f = 0$, so $C_S'^f$ is the S -value of a cuspidal 1-cocycle and admits Brown's decomposition

$$C_S'^f = \omega^+(f) P_S^+ + \omega^-(f) P_S^- + c(f) (X^{10} - Y^{10}), \quad (12)$$

with the $c(f)$ ambiguity absorbing the $p_{10} = 0$ convention.

4 The universal polynomial kernels

Definitions 2.1, 3.3 and Proposition 3.2 express $C_S'^f$ as an explicit $\mathbb{Q}$ -linear combination of the moments $\{N_j, M_j\}_{j=0}^{10}$ of f . Reading off interior-monomial coefficients $[C_S'^f]_{X^{10-j}Y^j}$ from (12) and dividing by c_S^j gives $\omega^{\varepsilon_j}(f)$ as a $\mathbb{Q}$ -linear functional of the moments. Each N_j (resp. M_j) is an integral of f against τ^j on the S -segment (resp. T -segment), so chaining gives an integral of f against an explicit polynomial in τ .

Theorem 4.1 (Universal kernel polynomials). *For each interior $j \in \{1, 2, \dots, 9\}$, define*

$$k_S^j(\tau) := (-1)^j \binom{10}{j} \tau^j, \quad (13)$$

$$k_T^j(\tau) := \sum_{r=1}^{10} [\beta_{j,r} - (-1)^j \beta_{10-j,r}] \tau^r, \quad (14)$$

where $\beta_{l,r} \in \mathbb{Q}$ are the Bernoulli-operator weights from (10) computing the $X^{10-l}Y^l$ coefficient of P from the moment M_r ,

$$\beta_{l,r} = \frac{(-1)^r \binom{10}{r} B_{l-r+1} (10-r)!}{(l-r+1)! (10-l)!} \quad \text{for } 0 \leq l \leq 9, 1 \leq r \leq l+1, \quad (15)$$

and $\beta_{l,r} = 0$ otherwise (in particular $\beta_{10,r} = 0$ for all r , encoding $p_{10} = 0$). With these kernels, identity (2) of Theorem 1.1 holds for each interior j .

Proof. By Definition 3.3, $[C_S'^f]_{X^{10-j}Y^j} = [C_S^f]_{X^{10-j}Y^j} - [P(Y, -X) - P(X, Y)]_{X^{10-j}Y^j}$. The first term is $(2\pi i)^{11} (-1)^j \binom{10}{j} N_j$ by (7); since $N_j = \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^j d\tau$, the S -segment contribution to $\omega^{\varepsilon_j}(f) = [C_S'^f]_{X^{10-j}Y^j} / c_S^j$ is $(2\pi i)^{11} / c_S^j \cdot \int_{-1/\tau_0}^{\tau_0} f(\tau) k_S^j(\tau) d\tau$ with k_S^j as in (13).

For the second term, write $P = \sum_l p_l X^{10-l} Y^l$ and $P(Y, -X) = \sum_l p_l (-1)^l X^l Y^{10-l}$. The $X^{10-j}Y^j$ coefficient of $P(Y, -X)$ is $(-1)^{10-j} p_{10-j} = (-1)^j p_{10-j}$ (since $(-1)^{10} = 1$), so $[P(Y, -X) - P(X, Y)]_{X^{10-j}Y^j} = (-1)^j p_{10-j} - p_j$. Each p_l is a $\mathbb{Q}$ -linear combination of moments $p_l = (2\pi i)^{11} \sum_r \beta_{l,r} M_r$ via (10); direct expansion gives the formula (15). Therefore

$$[P(Y, -X) - P(X, Y)]_{X^{10-j}Y^j} = (2\pi i)^{11} \sum_{r=1}^{10} [(-1)^j \beta_{10-j,r} - \beta_{j,r}] M_r,$$

and the T -segment contribution to $\omega^{\varepsilon_j}(f)$ is $-(2\pi i)^{11} / c_S^j \cdot \int_{\tau_0}^{-1/\tau_0} f(\tau) Q(\tau) d\tau$ where $Q(\tau) = \sum_r [(-1)^j \beta_{10-j,r} - \beta_{j,r}] \tau^r$. The negative sign flips this into $k_T^j(\tau)$ as in (14). Adding the S - and T -contributions gives (2). Independence of τ_0 and j is Theorem 6.1 below. $\square$

The kernels are universal: they depend only on j , not on f or τ_0 . Computing them via sympy and the Bernoulli formula (10)–(15) [12] gives Table 1.

Remark 4.2. *The S -kernel $k_S^j(\tau)$ is a pure monomial of degree j — it comes directly from the bare τ^j moment N_j . All complexity sits in the T -kernel $k_T^j(\tau)$, which encodes the Bernoulli-rational coboundary correction.*

j	ε_j	c_S^j	$k_S^j(\tau)$	$k_T^j(\tau)$
1	−	4	-10τ	$-5\tau + \frac{7}{2}\tau^2 + 5\tau^4 - 7\tau^6 + \frac{15}{2}\tau^8 - 5\tau^9 + \tau^{10}$
2	+	1	$45\tau^2$	$-9\tau + \frac{45}{2}\tau^2 - 5\tau^3 - 21\tau^5 + 30\tau^7 - \frac{45}{2}\tau^8 + 5\tau^9$
3	−	−25	$-120\tau^3$	$40\tau^2 - 60\tau^3 - 5\tau^4 + 70\tau^6 - 60\tau^7 + 15\tau^8$
4	+	−3	$210\tau^4$	$12\tau - 105\tau^3 + 105\tau^4 + 63\tau^5 - 105\tau^6 + 30\tau^7$
5	−	42	$-252\tau^5$	$-42\tau^2 + 210\tau^4 - 252\tau^5 + 84\tau^6$
6	+	3	$210\tau^6$	$-12\tau + 105\tau^3 - 105\tau^4 - 63\tau^5 + 105\tau^6 - 30\tau^7$
7	−	−25	$-120\tau^7$	$40\tau^2 - 60\tau^3 - 5\tau^4 + 70\tau^6 - 60\tau^7 + 15\tau^8$
8	+	−1	$45\tau^8$	$9\tau - \frac{45}{2}\tau^2 + 5\tau^3 + 21\tau^5 - 30\tau^7 + \frac{45}{2}\tau^8 - 5\tau^9$
9	−	4	$-10\tau^9$	$-5\tau + \frac{7}{2}\tau^2 + 5\tau^4 - 7\tau^6 + \frac{15}{2}\tau^8 - 5\tau^9 + \tau^{10}$

Table 1: The nine universal polynomial-kernel pairs (k_S^j, k_T^j) of Theorem 4.1, with Brown’s basis values $c_S^j = [P_S^{\varepsilon_j}]_{X^{10-j}Y^j}$. Computed exactly via δ_T^{-1} in sympy rationals [12]. Pairs (1, 9) and (3, 7) share k_T identically; pairs (2, 8) and (4, 6) share k_T up to overall sign (matching the sign flip in c_S^j); $j = 5$ is self-dual.

5 Specialisation to Δ and $\widehat{\Delta}$

We now specialise Theorem 1.1 to $f = \Delta$ and $f = \widehat{\Delta}$ and verify Corollaries 1.3, 1.4 numerically.

Corollary 5.1 (Explicit formula for $\omega^\pm(\Delta)$). *For $\Delta \in S_{12}$, any $\tau_0 \in \mathbb{H}$, and any interior $j \in \{1, \dots, 9\}$,*

$$\omega^{\varepsilon_j}(\Delta) = \frac{(2\pi i)^{11}}{c_S^j} \left[\int_{-1/\tau_0}^{\tau_0} \Delta(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} \Delta(\tau) k_T^j(\tau) d\tau \right]. \quad (16)$$

Corollary 5.2 (Explicit formula for $\eta^\pm(\widehat{\Delta})$). *For $\widehat{\Delta} \in S_{12}^!$, any $\tau_0 \in \mathbb{H}$, and any interior $j \in \{1, \dots, 9\}$,*

$$\eta^{\varepsilon_j}(\widehat{\Delta}) = \frac{(2\pi i)^{11}}{c_S^j} \left[\int_{-1/\tau_0}^{\tau_0} \widehat{\Delta}(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} \widehat{\Delta}(\tau) k_T^j(\tau) d\tau \right]. \quad (17)$$

The kernels in (16) and (17) are *identical*; only the modular form f under the integral changes.

Numerical verification. At $\tau_0 = 0.3 + 1.2i$, the integrals in (16) and (17) are computed at 45-digit `mpmath` precision in [9]. Defining the DR-regularised L -value $L_{\text{reg}}(f, j+1)$ by $\omega^{\varepsilon_j}(f)c_S^j/[(2\pi)^{10-j}(-i)^j \binom{10}{j} j!]$ (which agrees with the classical $L(\Delta, j+1)$ for $f = \Delta$, by Lemma 5.3 below):

j	ε_j	$L_{\text{reg}}(\Delta, j+1)$	extracted $\omega^{\varepsilon_j}(\Delta)$
1	−	0.14637454209126598...	−5 585 015.379310401 ... i
2	+	0.31524156588099308...	−68 916 772.80959519 ...
3	−	0.50161764751090221...	−5 585 015.379310401 ... i
4	+	0.66670918843400364...	−68 916 772.80959519 ...
5	−	0.79212283864603056...	−5 585 015.379310401 ... i
6	+	0.87735412538866091...	−68 916 772.80959519 ...
7	−	0.93070703029812609...	−5 585 015.379310401 ... i
8	+	0.96212645969442586...	−68 916 772.80959519 ...
9	−	0.97980908825122051...	−5 585 015.379310401 ... i

All nine rows lockstep to relative residual $< 6 \times 10^{-42}$ [11]. The same nine integrals with $\hat{\Delta}$ in place of Δ produce η^+, η^- in lockstep to relative residual $\sim 10^{-17}$ [13], agreeing with Brown's reported values to all 20 reported digits (Corollary 1.4); the slightly reduced precision reflects the larger absolute magnitudes ($|\eta^\pm| \sim 10^{11}$ versus $|\omega^\pm| \sim 10^7$).

Lemma 5.3 (Mellin-transform recovery for Δ). *As $\tau_0 \rightarrow i\infty$, the T -segment integral in (16) vanishes (since all $M_j(\Delta, \tau_0) \rightarrow 0$ by cuspidality of Δ), the S -segment becomes $[0, i\infty]$, and the Mellin transform $\int_0^{i\infty} \Delta(\tau) \tau^j d\tau = i^{j+1} j! / (2\pi)^{j+1} \cdot L(\Delta, j+1)$ identifies $L_{\text{reg}}(\Delta, j+1) = L(\Delta, j+1)$ with the standard critical L -value. (16) then collapses to the classical Eichler–Shimura formula $\omega^{\varepsilon_j}(\Delta) = (2\pi)^{10-j} (-i)^j \binom{10}{j} j! L(\Delta, j+1) / c_S^j$.*

Proof. $M_j(\Delta, \tau_0) \rightarrow 0$ exponentially because the T -path integrand $\Delta(\tau) \tau^j$ decays as $|q|^1 = e^{-2\pi \text{Im}(\tau)}$ on the segment. Hence $P \rightarrow 0$ by Proposition 3.2, the T -segment contribution to (16) drops to zero, and the S -segment $[-1/\tau_0, \tau_0]$ stretches to $[0, i\infty]$. The Mellin identification then follows from the Hecke–Weil integral representation of $L(\Delta, s)$. $\square$

For $\hat{\Delta}$ the $\tau_0 \rightarrow i\infty$ limit is unavailable — both integrals individually diverge as the q^{-1} Fourier mode amplifies each $|q_0|^{-1} = e^{2\pi \text{Im}(\tau_0)} \rightarrow \infty$. But for finite $\tau_0 \in \mathbb{H}$ both integrals are finite and (17) yields $\eta^\pm(\hat{\Delta})$. The basepoint is the regulator.

Remark 5.4 (Exponentially convergent q -series expansion). *Substituting the Fourier expansion $f(\tau) = \sum_{n \geq -N_f} a_f(n) e^{2\pi i n \tau}$ into (16) (resp. (17)) and integrating term-by-term via the elementary identity $\int e^{2\pi i n \tau} \tau^j d\tau = \sum_{k=0}^j \frac{(-1)^k j!}{(j-k)!(2\pi i n)^{k+1}} e^{2\pi i n \tau} \tau^{j-k}$ gives an exponentially convergent series for the periods:*

$$\omega^{\varepsilon_j}(f) = \frac{(2\pi i)^{11}}{c_S^j} \sum_{n \neq 0} a_f(n) \mathcal{K}^j(n, \tau_0) + (\text{constant-term piece, vanishing for } f \in S_k^!), \quad (18)$$

where $\mathcal{K}^j(n, \tau_0)$ is an explicit elementary function of n and τ_0 built from finite polynomial combinations of $q_0^n = e^{2\pi i n \tau_0}$, $\tilde{q}_0^n = e^{-2\pi i n / \tau_0}$, τ_0^a , $(\tau_0 - 1)^a$, $(-1/\tau_0)^a$, and inverse powers $1/(2\pi i n)^{b+1}$. Since $|q_0|, |\tilde{q}_0| < 1$ for every $\tau_0 \in \mathbb{H}$, the series converges super-exponentially in $|n|$.

For $f = \Delta$, only $n \geq 1$ contributes with $a_f(n) = \tau(n)$, yielding an exponentially convergent series for the classical $\omega^\pm(\Delta)$ — the explicit linear functional of $\{\tau(n)\}_{n \geq 1}$ promised in the introduction. For $f = \hat{\Delta}$, the $n = -1$ term contributes one extra finite piece $q_0^{-1} \mathcal{K}^j(-1, \tau_0)$; the $n \geq 2$ terms decay super-exponentially in n despite the $a_{\hat{\Delta}}(n) \sim e^{4\pi\sqrt{n}}$ coefficient growth. Comparing the Δ -version of (18) to BFK's incomplete-gamma expansion of $L(\Delta, s)$ [3] provides a concrete sanity check; writing the same expansion for $\eta^\pm(\hat{\Delta})$ produces the natural DR-analog of BFK's regularised L -value series, valid uniformly across $S_{12}^!$. Detailed term-by-term comparison of these series is deferred to a follow-up note.

6 Cohomology and basepoint independence

Theorem 6.1 (Cohomological well-definition). *The class $[C^f] \in H^1(\Gamma; V_{10} \otimes \mathbb{C})$ is independent of τ_0 . For $f \in S_{12}^!$, this class is cuspidal, and Brown's decomposition (12) reads off cohomologically-meaningful coordinates: $\omega^+(f) = \text{coordinate of } [C^f] \text{ in } [P^+]$, $\omega^-(f) = \text{coordinate in } [P^-]$.*

Proof. For $\tau_0, \tau_1 \in \mathbb{H}$, define the basepoint-shift potential

$$h(X, Y; \tau_0, \tau_1) := (2\pi i)^{11} \int_{\tau_0}^{\tau_1} f(\tau) (X - \tau Y)^{10} d\tau \in V_{10} \otimes \mathbb{C}. \quad (19)$$

The same path-additivity / modularity calculation as in Lemma 2.2, applied to the basepoint-shifted integrals, gives

$$C_\gamma^f(\tau_1) - C_\gamma^f(\tau_0) = -[h(\tau_0, \tau_1)|_\gamma - h(\tau_0, \tau_1)], \quad (20)$$

manifestly a coboundary in $V_{10} \otimes \mathbb{C}$. Hence $[C_\bullet^f]$ is independent of τ_0 . Cuspidality of the class for $f \in S_{12}^!$ follows from Proposition 3.1. The basis-coordinate identification is (12). $\square$

Corollary 6.2 (τ_0 - and j -independence in Theorem 1.1). *The right side of (2) is independent of $\tau_0 \in \mathbb{H}$ (Theorem 6.1) and of the choice of interior $j \in \{1, \dots, 9\}$ within its parity sector (since all interior- j extractions read off the same basis coordinate of $[C^f]$ in (12)). Together with Theorem 4.1, this completes the proof of Theorem 1.1. Numerically, $|\Delta\omega^\pm| < 3 \times 10^{-38}$ and $|\Delta\eta^\pm| < 1.3 \times 10^{-32}$ between $\tau_0 = 0.3 + 1.2i$ and $\tau_0 = -0.2 + 0.9i$ [9].*

Remark 6.3 ($C_S'^f$ in Zagier's basis). *Using (12) together with $P_S^+ = -\frac{36}{691}P_0 + P_1$, $P_S^- = P_2$:*

$$C_S'^f = a_0(f)P_0 + \omega^+(f)P_1 + \omega^-(f)P_2,$$

with $a_0(f) = c(f) - \frac{36}{691}\omega^+(f) = [C_S'^f]_{X^{10}}$ the convention-dependent Eisenstein coordinate. At $\tau_0 = 0.3 + 1.2i$ [13]:

$$\begin{aligned} C_S'^\Delta &= (2.752 \times 10^5 - 6.156 \times 10^5 i)P_0 + (-68\,916\,772.80959519 \dots)P_1 + (-5\,585\,015.379310402 \dots i)P_2, \\ C_S'^{\hat{\Delta}} &= (-1.532 \times 10^{11} + 6.163 \times 10^{11} i)P_0 + (127\,202\,100\,647.1771 \dots)P_1 + (10\,276\,732\,343.64913 \dots i)P_2. \end{aligned}$$

The P_1, P_2 coordinates are precisely $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$.

7 Comparison with the BFK regularisation

Bringmann–Fricke–Kent [3] regularise the divergent integral $\int_0^{i\infty} f(\tau) \tau^{s-1} d\tau$ for weakly holomorphic $f \in S_k^!$ via a non-holomorphic exponential factor $e^{2\pi\nu \text{Im}(\tau)}$ continued in ν . At integer $s \in \{1, \dots, k-1\}$ their regularised L^* -value reduces to the absolutely-convergent incomplete-gamma sum

$$L_{\text{BFK}}^*(f, s) = \sum_{n \neq 0} a_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (21)$$

where the sum extends over *all* nonzero Fourier modes, including the negative-index principal-part modes of weakly holomorphic forms. At integer s the incomplete-gamma function reduces to a polynomial in its second argument ($\Gamma(s, x) = (s-1)! e^{-x} \sum_{r < s} x^r / r!$), so each $n < 0$ contribution $a_f(n) \Gamma(s, -2\pi|n|) / (-2\pi|n|)^s$ is a finite complex number and (21) is a well-defined finite sum. Zagier's formula then assembles these eleven L^* -values into a polynomial $r_{\text{BFK}}(f) \in V_{k-2} \otimes \mathbb{C}$.

Proposition 7.1 (r_{BFK} and the DR-B representative agree). *Let $f \in \{\Delta, \widehat{\Delta}\}$. Then:*

1. $r_{\text{BFK}}(f) \in W_{12}$: both period relations $r_{\text{BFK}}(f)|(1+S) = 0$ and $r_{\text{BFK}}(f)|(1+U+U^2) = 0$ (with $U = TS$, weight -10 slash action) hold to relative residual $\sim 10^{-48}$ at 50-digit *mpmath* precision [14].
2. The polynomial- V_{10} DR-B representative $C_S^{f'}(\tau_0)/(2\pi i)^{11}$ of Theorem 1.1 agrees with $r_{\text{BFK}}(f)$ exactly on the nine interior monomials $X^1, \dots, X^9$ (relative residual $< 10^{-44}$). At the boundary monomials X^0, X^{10} the two polynomials differ by an Eisenstein-direction term $c(\tau_0)(1 - X^{10})$; the constant c is τ_0 -dependent (reflecting the boundary freedom in the polynomial- V_{10} DR-B representative noted after $(\star)$) and at $\tau_0 = 0.3 + 1.2i$ has magnitude $\approx 5.6 \times 10^{-3}$ for $f = \Delta$ and $\approx 1.0 \times 10^3$ for $f = \widehat{\Delta}$.
3. The Hurwitz-zeta-extended DR-B representative $r_{\text{DR}}^{\widetilde{k}_T}(f)$ of §8 agrees with $r_{\text{BFK}}(f)$ at all eleven monomials $X^0, \dots, X^{10}$ to relative residual $\sim 10^{-45}$, for both $f = \Delta$ and $f = \widehat{\Delta}$. Equivalently, $\Lambda_{\text{BFK}}^*(f, s) = \Lambda_{\text{DR}}^*(f, s)$ at all integer $s \in \{1, \dots, 11\}$, and (numerically) at the non-integer test points $s \in \{-0.3, 0.5, 1.5, \dots, 6.5\}$ and complex test points $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$ for both f , at relative residual $\sim 10^{-47}$ [14]. The Hurwitz-zeta T -kernel is the DR-B representative that pins the boundary coefficients down τ_0 -canonically.

What distinguishes the two constructions, then, is not the value of the period polynomial they deliver (they deliver the same one) but their geometry. BFK uses two infinite contours $\int_0^{i\infty}$ and $(\int_0^{i\infty})|S$ with a non-holomorphic regulator inserted at both endpoints; DR-B uses two *finite* contours $\{[-1/\tau_0, \tau_0], [\tau_0 - 1, \tau_0]\}$ that stay strictly inside $\mathbb{H}$ and require no regulator. The finite-contour construction is what enables the natural extension to meromorphic forms with interior poles (§9): a regulator-based scheme has no immediate analog there because the cusp-exchange symmetry that pins down BFK's regulator at the cusps has no analog at an interior pole.

8 The complex- s extension: a finite-contour L -function via Hurwitz zeta

Theorem 1.1 extracts the period-polynomial coefficients of $f \in S_{12}^1$ at the nine interior integer monomials $j \in \{1, \dots, 9\}$ via finite-contour integrals on the DR-B contour pair $\{[-1/\tau_0, \tau_0], [\tau_0 - 1, \tau_0]\}$. For holomorphic $f = \Delta$ we now extend the same construction to general complex s , producing a closed-form expression for the completed L -function $\Lambda^*(\Delta, s) := (2\pi)^{-s}\Gamma(s)L(\Delta, s)$ as an integral pair over the *same* DR-B contour, with the polynomial T -kernel k_T^j of Table 1 replaced by a Hurwitz-zeta function of τ .

The gap identity. For $\Delta \in S_{12}$ and any $\tau_0 \in \mathbb{H}$, splitting the canonical Mellin integral $\int_0^{i\infty} \Delta(\tau) \tau^{s-1} d\tau$ at the DR-B endpoints $\{-1/\tau_0, \tau_0\}$ and using S -modularity of Δ on the $[0, -1/\tau_0]$ piece (via $\tau \mapsto -1/\tau$) gives, for all $s \in \mathbb{C}$,

$$\Lambda^*(\Delta, s) = i^{-s} \int_{-1/\tau_0}^{\tau_0} \Delta(\tau) \tau^{s-1} d\tau + \sum_{n \geq 1} \tau(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(12-s, -2\pi i n \tau_0)}{(2\pi n)^{12-s}} \right]. \quad (22)$$

The second piece is the BFK incomplete-gamma sum of [3] *split at τ_0 rather than at $\tau = i$* : at $\tau_0 = i$ the argument $-2\pi i n \tau_0 = 2\pi n$ is real and (22) reduces to the standard BFK formula. Both pieces on the right of (22) are individually entire in s and depend on τ_0 , but their sum equals the τ_0 -invariant $\Lambda^*(\Delta, s)$. Identity (22) is verified numerically to relative residual $\sim 10^{-54}$ at $\tau_0 = 0.3 + 1.2i$, 45-digit precision, for $s = 1, \dots, 11$ [14].

The τ_0 -invariance condition. We seek a kernel $\tilde{k}_T(\tau, s)$ such that the BFK-split-at- τ_0 sum in (22) is replaced by a finite-contour integral $\int_{\tau_0-1}^{\tau_0} \Delta(\tau) \tilde{k}_T(\tau, s) d\tau$ over the same T -segment used in Theorem 1.1. Differentiating the candidate

$$\Phi_s(\tau_0) := i^{-s} \left[\int_{-1/\tau_0}^{\tau_0} \Delta(\tau) \tau^{s-1} d\tau + \int_{\tau_0-1}^{\tau_0} \Delta(\tau) \tilde{k}_T(\tau, s) d\tau \right]$$

in τ_0 , using $\Delta(-1/\tau_0) = \tau_0^{12} \Delta(\tau_0)$ for the S -segment endpoint and $\Delta(\tau_0 - 1) = \Delta(\tau_0)$ for the T -segment endpoint, the requirement $d\Phi_s/d\tau_0 \equiv 0$ reduces (canceling the common factor $i^{-s} \Delta(\tau_0)$) to

$$\tilde{k}_T(\tau, s) - \tilde{k}_T(\tau - 1, s) = -\tau^{s-1} + e^{i\pi(s-1)} \tau^{11-s} \quad \text{for all } \tau \in \mathbb{H}. \quad (23)$$

This is the complex- s analog of the polynomial coboundary equation ($\star$): it asks for an inverse of δ_T on the two-term right-hand side.

Hurwitz zeta solves (23). The Hurwitz raising identity $\zeta(\sigma, a) - \zeta(\sigma, a+1) = a^{-\sigma}$, applied at $(\sigma, a) = (1-s, \tau)$ and $(s-11, \tau)$, gives

$$\zeta(1-s, \tau+1) - \zeta(1-s, \tau) = -\tau^{s-1}, \quad \zeta(s-11, \tau+1) - \zeta(s-11, \tau) = -\tau^{11-s}.$$

Combining with the sign $e^{i\pi(s-1)}$ from (23):

Theorem 8.1 (Complex- s finite-contour L -function for Δ). *For all $s \in \mathbb{C} \setminus \{1, 12\}$ and all $\tau_0 \in \mathbb{H}$,*

$$\Lambda^*(\Delta, s) = i^{-s} \left[\int_{-1/\tau_0}^{\tau_0} \Delta(\tau) \tau^{s-1} d\tau + \int_{\tau_0-1}^{\tau_0} \Delta(\tau) \tilde{k}_T(\tau, s) d\tau \right], \quad (24)$$

where the Hurwitz-zeta T -kernel is

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-11, \tau+1). \quad (25)$$

Both contour integrals are finite for every $\tau_0 \in \mathbb{H}$, with no regularisation, analytic continuation, or limit required. At integer $s = j+1 \in \{1, 2, \dots, 11\}$, both ζ -terms in (25) collapse to Bernoulli polynomials in τ via the standard identity $\zeta(-n, a) = -B_{n+1}(a)/(n+1)$, and $\tilde{k}_T(\tau, j+1)$ becomes a polynomial in τ . For the nine interior values $j \in \{1, \dots, 9\}$ this polynomial agrees (up to the trivial rescaling $(-1)^j \binom{10}{j}$ that converts the $\omega^\pm$ -normalised kernel to the L -value-normalised one) with the polynomial T -kernel k_T^j of Table 1. The two boundary cases $s = 1$ and $s = 11$ — which the bounded-degree V_{10} Bernoulli inversion of §3 cannot fix (the convention $p_{10} = 0$ leaves an unresolved Eisenstein direction there) — are recovered seamlessly in (24) because the Hurwitz inversion does not need a polynomial-degree truncation.

Proof. Both sides of (24) are meromorphic in s , with poles on the right side only from the explicit Hurwitz factors at $s = 1, 12$ (poles cancelled, in the sum with the S -segment, by the analytic structure of Λ^*). By construction (23), the right side of (24) is τ_0 -invariant: differentiating in τ_0 reduces to identity (23), which holds for $\tilde{k}_T$ from (25) by the Hurwitz raising identity applied to each ζ term separately. Hence the right side equals its $\tau_0 \rightarrow i\infty$ limit: the T -segment integral $\int_{\tau_0-1}^{\tau_0} \Delta \tilde{k}_T d\tau$ vanishes exponentially because Δ does and $\tilde{k}_T$ grows at most algebraically in τ , while the S -segment integral $i^{-s} \int_{-1/\tau_0}^{\tau_0} \Delta \tau^{s-1} d\tau$ approaches $i^{-s} \int_0^{i\infty} \Delta \tau^{s-1} d\tau = \Lambda^*(\Delta, s)$ by the Hecke–Weil Mellin identity (extending Lemma 5.3 from integer j to complex s). This identifies the right side with $\Lambda^*(\Delta, s)$. $\square$

Numerical verification. Formula (24) reproduces the standard BFK-at- $\tau = i$ incomplete-gamma value of $\Lambda^*(\Delta, s)$ to relative residual $\sim 10^{-53}$ at 45-digit `mpmath` precision [14], at every integer $s \in \{1, 2, \dots, 11\}$ (including the two boundaries $s = 1, 11$ unreachable by Theorem 1.1), every half-integer test point $s \in \{-0.3, 0.5, 1.5, 2.5, \dots, 6.5\}$, and every complex test point $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$. Basepoint-independence is verified directly: at fixed $s = 3 + 1.5i$, evaluating the right side of (24) at $\tau_0 \in \{0.3 + 1.2i, 1.5i, -0.4 + 0.8i, 0.2 + 2i\}$ gives one value to relative residual $\sim 10^{-53}$.

Structural interpretation. The polynomial T -kernels k_T^j of Theorem 4.1 are the integer- s specialisations of $\tilde{k}_T(\tau, s)$. The bounded-degree Bernoulli inversion on V_{10} in §3 is the integer-weight shadow of the analytic Hurwitz raising identity

$$\delta_T[\zeta(1-s, \tau+1)] = \zeta(1-s, \tau+2) - \zeta(1-s, \tau+1) = -(\tau+1)^{s-1},$$

which inverts δ_T uniformly at all complex s without bounding the polynomial degree. In cohomological language, the finite-dimensional coefficient system V_n of Eichler–Shimura extends to an analytic family of principal-series-type coefficient systems on $\mathbb{H}$ parametrised by s , and Theorem 8.1 is the explicit DR-B representative for Δ in this larger system. This is the holomorphic, finite-contour analog of the Lewis–Zagier period-function framework for Maass wave forms [6].

Same formula for Δ and $\hat{\Delta}$ at all s . The right side of (24) is finite for any $f \in S_{12}^!$ in place of Δ , and indeed for any meromorphic modular form of weight 12 with no pole on the DR-B contour pair — both segments stay strictly inside $\mathbb{H}$, and the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ is analytic in $\tau \in \mathbb{H}$. Applying (24) to $\hat{\Delta}$ produces $\Lambda_{\text{DR}}^*(\hat{\Delta}, s)$ for every $s \in \mathbb{C}$; at integer $s \in \{1, \dots, 11\}$ this equals the BFK L^* -value of (21) (Proposition 7.1), so that $L^{\text{DR}}(\hat{\Delta}, s) = L_{\text{BFK}}^*(\hat{\Delta}, s)$ at all eleven integer points including the boundaries $s = 1, 11$. Equation (24) thereby gives the canonical extension of $L_{\text{BFK}}^*(\hat{\Delta}, \cdot)$ to all $s \in \mathbb{C}$, as a finite-contour integral pair with no regulator and no analytic continuation step. The remaining analytic question — whether this extension agrees with $L_{\text{BFK}}^*(\hat{\Delta}, \cdot)$ at *all* complex s and not only at the eleven integer points — is addressed in §8.1 below via a basepoint specialisation that turns (24) into a single integral on a compact contour. The analog of the functional equation $\Lambda^*(s) = i^k \Lambda^*(k-s)$ in finite-contour form is a closely related next step.

8.1 The $\tau_0 = i$ specialisation: BFK = DR-B at all complex s (sketch)

The right side of (24) is τ_0 -invariant for $f \in S_{12}^!$, so we are free to pick a basepoint that simplifies the contour pair maximally. The natural choice is $\tau_0 = i$: since $S \cdot i = -1/i = i$, the elliptic fixed point of S on $\mathbb{H}$, the S -segment $[-1/\tau_0, \tau_0] = [i, i]$ collapses to a single point. Equation (24) reduces to

$$\Lambda_{\text{DR}}^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau, \quad (26)$$

a *single* integral over the compact T -segment $[i-1, i]$. This is the dual specialisation to the classical Mellin choice $\tau_0 \rightarrow i\infty$: at the cusp basepoint the T -segment dies exponentially and the S -segment limit recovers the Mellin contour $[0, i\infty]$; at the elliptic-fixed-point basepoint the S -segment vanishes identically and the T -segment survives. Both specialisations exploit a fixed point of $Y(1) = \Gamma \backslash \mathbb{H}$, just different ones.

Basepoint	Vanishing segment	Surviving contour	What it gives
$\tau_0 \rightarrow i\infty$ (cusp)	T (exponentially)	S -segment $\rightarrow [0, i\infty]$	Classical Mellin
$\tau_0 = i$ (S -fixed point)	S (identically)	T -segment $[i-1, i]$	Elliptic-fixed-point form

Mode-by-mode evaluation at integer s . Apply (26) to $f = \widehat{\Delta} = q^{-1} + \sum_{n \geq 2} a(n) q^n$. The q -series converges uniformly on the compact segment $[i-1, i]$ (the polar mode satisfies $|q^{-1}| \leq e^{2\pi}$ there), and $\widetilde{k}_T(\tau, s)$ is continuous in τ , so sum and integral commute:

$$\Lambda_{\text{DR}}^*(\widehat{\Delta}, s) = i^{-s} \sum_n a(n) \int_{i-1}^i e^{2\pi i n \tau} \widetilde{k}_T(\tau, s) d\tau. \quad (27)$$

At integer $s \in \{1, \dots, 11\}$, both halves of $\widetilde{k}_T$ collapse to Bernoulli polynomials in $\tau + 1$ via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$. The per-mode integral becomes a finite sum of $\int_i^{i+1} e^{2\pi i n \sigma} \sigma^m d\sigma$ terms (substituting $\sigma = \tau + 1$), each of which evaluates in closed form by repeated integration by parts:

$$\int_i^{i+1} e^{2\pi i n \sigma} \sigma^m d\sigma = \sum_{k=0}^m \frac{(-1)^k m!}{(m-k)! (2\pi i n)^{k+1}} [(i+1)^{m-k} - i^{m-k}] e^{-2\pi i n}, \quad (28)$$

using $e^{2\pi i n(i+1)} = e^{2\pi i n \cdot i} \cdot e^{2\pi i n} = e^{-2\pi i n} \cdot 1$ for integer n . For positive integer n , summing (28) against the Bernoulli-polynomial coefficients of the kernel reproduces the BFK incomplete-gamma summand $\Gamma(s, 2\pi n)/(2\pi n)^s + i^k \Gamma(k-s, 2\pi n)/(2\pi n)^{k-s}$ directly, via the explicit polynomial expression

$$\Gamma(N, x) = (N-1)! e^{-x} \sum_{m=0}^{N-1} \frac{x^m}{m!} \quad (N \in \mathbb{Z}_{\geq 1}).$$

For the polar mode $n = -1$, the *same* closed-form expression (28) matches $\Gamma(N, -2\pi)/(-2\pi)^N$ via the same polynomial identity, which is the analytic-continuation value of the upper incomplete gamma (entire in its second argument). The polar mode is therefore handled by the cuspidal IBP machinery without any separate regularisation step.

Removable kernel pole at $s = 12$ (and $s = 0$). The kernel $\widetilde{k}_T(\tau, s)$ has an apparent simple pole at $s = 12$ from $\zeta(s-11, \tau+1) = \zeta(1, \tau+1)$ (and at $s = 0$ from $\zeta(1, \tau+1)$ in the first half), but the residue contribution to every per-mode integral vanishes identically:

$$\int_{i-1}^i e^{2\pi i n \tau} d\tau = 0 \quad \text{for every nonzero integer } n,$$

because $e^{2\pi i n(i-1)} = e^{2\pi i n \cdot i} e^{-2\pi i n} = e^{2\pi i n \cdot i}$. The per-mode DR-B integral is therefore entire in s , consistent with the BFK side being entire.

Numerical verification. At 50-digit `mpmath` precision [16, 17], per-mode equality of the right side of (27) with the BFK incomplete-gamma value holds at every tested (n, s) :

s	$n = -1$	$n = +1$	$n = +2$	$n = +3$
1	10^{-51}	10^{-53}	10^{-57}	10^{-59}
3	10^{-58}	10^{-53}	10^{-57}	10^{-59}
6	10^{-51}	10^{-54}	10^{-57}	10^{-59}
9	10^{-56}	10^{-53}	10^{-57}	10^{-59}
11	10^{-51}	10^{-53}	10^{-57}	10^{-59}

Values shown are relative residuals between the per-mode DR-B integral and the per-mode BFK incomplete-gamma value. The boundary values $s = 1, 11$ are matched without modification (no V_{10} -degree truncation enters). Modes $s = 3$ and $s = 9$ give identical per-mode values, reflecting the $s \leftrightarrow k - s$ symmetry through $i^k = 1$.

From Hurwitz to incomplete gamma: the per-mode identity at all complex s . The IBP closed form (28) above evaluates each per-mode integral at integer s , where the Hurwitz-zeta kernel collapses to a polynomial in τ . A standard unfolding identity reduces the per-mode integral to an incomplete gamma directly, with no integer- s restriction, eliminating any appeal to Carlson interpolation.

Lemma 8.2 (Hurwitz unfolding). *For every positive integer n and every $\nu \in \mathbb{C}$ with $\text{Re}(\nu) > 1$,*

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (29)$$

Both sides are entire functions of ν , so (29) holds in fact for every $\nu \in \mathbb{C}$ by analytic continuation in ν .

Proof. For $\text{Re}(\nu) > 1$ the Hurwitz series $\zeta(\nu, \sigma) = \sum_{m \geq 0} (\sigma + m)^{-\nu}$ converges uniformly on the compact segment $[i, i+1]$ (cf. [15, §25.11]), so the sum and the integral may be exchanged. In the m -th summand substitute $u = \sigma + m$; the prefactor $e^{2\pi i n m} = 1$ for integer n removes all m -dependence except in the limits of integration, and the resulting integrals telescope:

$$\sum_{m \geq 0} \int_{i+m}^{i+m+1} e^{2\pi i n u} u^{-\nu} du = \int_i^{i+\infty} e^{2\pi i n u} u^{-\nu} du,$$

a single integral along the horizontal ray $\{i+t : t \geq 0\} \subset \mathbb{H}$. Next substitute $t = -2\pi i n u$, equivalently $u = it/(2\pi n)$, $du = i dt/(2\pi n)$, with $e^{2\pi i n u} = e^{-t}$ and principal-branch $u^{-\nu} = i^{-\nu} (2\pi n)^{\nu} t^{-\nu}$. The u -contour maps to the vertical line $\{2\pi n - is : s \geq 0\}$ in the right half t -plane. Since $e^{-t} t^{-\nu}$ is holomorphic in $\text{Re}(t) > 0$ and decays as $|t| \rightarrow \infty$ in this half-plane, Cauchy's theorem (applied to the rectangle $[2\pi n, R] - i[0, R]$, $R \rightarrow \infty$) deforms the vertical contour onto $[2\pi n, +\infty]$ along the positive real axis:

$$\int_{2\pi n}^{2\pi n - i\infty} e^{-t} t^{-\nu} dt = \int_{2\pi n}^{\infty} e^{-t} t^{-\nu} dt = \Gamma(1-\nu, 2\pi n)$$

by the integral definition of the upper incomplete gamma [15, §8.2]. Collecting Jacobian and branch factors yields (29) for $\text{Re}(\nu) > 1$. Both sides of (29) are entire in ν : the integral on the left is entire because $\zeta(\nu, \sigma)$ is meromorphic in ν with only a simple pole at $\nu = 1$ whose residue contribution to $\int_{i-1}^i e^{2\pi i n \tau} d\tau = 0$ vanishes for nonzero integer n (cf. the removable-pole observation below (28)), and the upper incomplete gamma $\Gamma(s, x)$ is entire in s for any fixed $x \neq 0$ [15, §8.2.7]. Analytic continuation in ν extends (29) to all complex ν . $\square$

The per-mode DR-B = BFK identity. Applying Lemma 8.2 to each half of the kernel (25) at $\nu = 1 - s$ and $\nu = s - 11$, and recalling the $e^{-2\pi i n} = 1$ identity used to shift the τ -integration from $[i-1, i]$ to $[i, i+1]$ (after $\sigma = \tau + 1$):

$$\begin{aligned} \int_{i-1}^i e^{2\pi i n \tau} \zeta(1-s, \tau+1) d\tau &= i^s (2\pi n)^{-s} \Gamma(s, 2\pi n), \\ \int_{i-1}^i e^{2\pi i n \tau} \zeta(s-11, \tau+1) d\tau &= i^{12-s} (2\pi n)^{s-12} \Gamma(12-s, 2\pi n). \end{aligned}$$

Combining with the relative phase $-e^{i\pi(s-1)}$ in $\tilde{k}_T$ and the overall i^{-s} prefactor of (26), the phases collapse: $i^{-s} \cdot i^s = 1$ in the first half; $-i^{-s} e^{i\pi(s-1)} i^{12-s} = -e^{i\pi(s-1)} i^{12-2s} = -e^{i\pi(s-1)} e^{-i\pi s} = -e^{-i\pi} = 1$ in the second. The per-mode DR-B value is therefore exactly

$$\Lambda_{\text{DR}}^{(n)}(s) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + \frac{\Gamma(12-s, 2\pi n)}{(2\pi n)^{12-s}}, \quad (30)$$

which is exactly the n -th summand of the BFK incomplete-gamma series (21) (using $i^{12} = 1$). Summing over $n \geq 1$ against the cuspidal Fourier coefficients of $\hat{\Delta}$ yields the full equality on the cuspidal part.

The polar mode $n = -1$. The same unfolding identity applies; the only change is that the t -contour after the substitution $t = -2\pi i n u = 2\pi i u$ runs from -2π upward to $-2\pi + i\infty$ rather than from $2\pi n$ downward. Deforming this contour to the positive real axis crosses the branch cut of $t^{-\nu}$ at the origin, and the resulting value of the integral is the analytic continuation of $\Gamma(1-\nu, x)$ in x from $x > 0$ to $x = -2\pi$. At every positive integer s in $\{1, \dots, 11\}$ the polynomial closed form

$$\Gamma(N, x) = (N-1)! e^{-x} \sum_{m=0}^{N-1} \frac{x^m}{m!} \quad (N \in \mathbb{Z}_{\geq 1}, x \in \mathbb{C})$$

makes the analytic continuation explicit and matches the IBP evaluation (28) verbatim; for non-integer s both sides are entire in s and equal by analytic continuation from any open neighbourhood of the integers. Equivalently, the polar mode contributes

$$\Lambda_{\text{DR}}^{(-1)}(s) = \frac{\Gamma(s, -2\pi)}{(-2\pi)^s} + \frac{\Gamma(12-s, -2\pi)}{(-2\pi)^{12-s}}$$

on a consistent branch, matching the BFK polar summand on the same branch.

Promotion to a theorem. Lemma 8.2 and the per-mode collapse (30), together with absolute convergence of the cuspidal Fourier series of $\hat{\Delta}$ on $[i-1, i]$ and the polar-mode discussion above, give the following upgrade of the earlier integer- s statement:

Theorem 8.3 (BFK = DR-B at all complex s). *For $\hat{\Delta}$, and by linearity in f for every $f \in S_{12}^!$,*

$$\Lambda_{\text{DR}}^*(f, s) = \Lambda_{\text{BFK}}^*(f, s) \quad \text{for every } s \in \mathbb{C},$$

where the left side is the finite-contour functional (24) (equivalently the single-segment (26) at $\tau_0 = i$) and the right side is the canonical entire extension of the BFK regulated L -function (21).

Theorem 8.3 eliminates the BFK regulator entirely in favour of a manifestly finite single-integral expression on the compact contour $[i-1, i] \subset \mathbb{H}$, with no regularisation, no asymptotic-subtraction argument, and no special case for the polar Fourier modes. The growth bound

$$|\zeta(\sigma, a)| = O(|\text{Im}(\sigma)|^{\max(0, 1/2 - \text{Re}(\sigma)) + \epsilon})$$

on vertical lines [15, §25.11.36] together with the entire-in- s Hurwitz-unfolding identity (29) renders the Carlson/Phragmén–Lindelöf interpolation argument unnecessary: equality holds as an exact analytic identity term by term in the Fourier expansion.

9 Outlook: meromorphic forms, and a gauge-invariance analogy

Recap. We have given (i) an explicit linear functional $f \mapsto (\omega^+(f), \omega^-(f))$ on $S_{12}^!$ via the finite-endpoint DR-B cocycle (Theorem 1.1); (ii) its extension to a finite-contour formula for the completed L -function $\Lambda^*(f, s)$ at every complex s via the Hurwitz-zeta T -kernel (Theorem 8.1); and (iii) the equality of this L -function with the BFK regularised L -function at all integer s for $f \in \{\Delta, \widehat{\Delta}\}$ (Proposition 7.1). The DR-B construction is therefore not a competitor to BFK; it is a manifestly finite reformulation of the same object, replacing BFK’s non-holomorphic regulator at the two cusps with two finite line segments inside $\mathbb{H}$.

Meromorphic forms with interior poles. The advantage of the finite-contour formulation appears once f acquires interior poles, where BFK has no analog: the cusp-exchange symmetry $\tau \rightarrow -1/\tau$ that pins BFK’s regulator at the two cusps has no counterpart at an interior $\tau_p \in \mathbb{H}$. The DR-B construction, by contrast, requires only that the contour pair $\{[-1/\tau_0, \tau_0], [\tau_0 - 1, \tau_0]\}$ avoid the poles of f . Choosing any τ_0 off the Γ -orbit of the polar locus $f^{-1}(\infty) \cap \mathbb{H}$, the formula (24) delivers $\Lambda_{\text{DR}}^*(f, s)$ for arbitrary meromorphic f of weight 12 at every $s \in \mathbb{C}$.

What changes is the τ_0 -dependence. For $f \in S_{12}^!$, $\Lambda_{\text{DR}}^*(f, s)$ is τ_0 -invariant by Theorem 8.1. For meromorphic f , the basepoint-shift identity (20) acquires residue contributions whenever the contour crosses a pole. Explicitly, for a simple pole of f at $\tau_p \in \mathbb{H}$, basepoint-shift across τ_p adds

$$2\pi i (2\pi i)^{11} \text{Res}(f, \tau_p) (X - \tau_p Y)^{10} \in V_{10} \otimes \mathbb{C} \quad (31)$$

to the cocycle polynomial C^f , and the analogous expression with derivatives in τ_p of $(X - \tau_p Y)^{10-i}$, $i = 0, \dots, m-1$, for a pole of order m . The presentation of $\Lambda_{\text{DR}}^*(f, s)$ as a finite-contour integral therefore becomes τ_0 -piecewise constant on the connected components of $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, with discrete *wall-crossing* jumps (31) at the Γ -orbits of the poles. This wall-crossing structure is a feature of the meromorphic L -function, not a defect of any one representative: it encodes the residue data of f in a form that is not visible in BFK’s regulator-based construction (because that construction cannot see interior poles in the first place).

Working out the explicit residue polynomials for canonical meromorphic forms (e.g. $1/E_6$ with simple poles at $\tau = i$ and its Γ -orbit; $1/E_4$ at $\tau = \rho$), and the arithmetic of the resulting L -functions and their wall-crossing data, is the natural successor program.

A gauge-invariance analogy, and an open question. The cocycle C^f carries more data than the cohomologically-meaningful periods $\omega^\pm(f)$: an entire coboundary direction δP must be subtracted, and the choice of representative P (parametrised by the unconstrained $p_{10} \in \mathbb{C}$) is irrelevant for the final answer. This carries strong echoes of off-shell gauge data in Feynman amplitudes with massless gauge bosons — longitudinal polarisations, Faddeev–Popov ghosts — which appear in the intermediate bookkeeping but cancel in physical (gauge-invariant) observables. In QFT, the on-shell S-matrix programme circumvents the gauge problem by building amplitudes directly from gauge-invariant on-shell building blocks (helicity variables, BCFW recursion, unitarity cuts [7, 8]), so the gauge-redundant intermediate data never enters: one is forced into “fix the gauge” only because one has made the upstream choice to work off-shell.

The analog question for periods: does there exist a “manifestly cohomological” construction that produces $\omega^\pm(f)$ directly from the Fourier coefficients of f , sidestepping the intermediate cocycle C^f and coboundary δP scaffolding entirely? The exponentially convergent q -coefficient series of Remark 5.4 is a partial step — input $\{a_f(n)\}$, output $\omega^\pm$, no C^f or P visible in the final formula — but the kernels appearing in that series were themselves derived via δ_T^{-1} , so the cocycle structure

remains implicit. A formulation in which the kernels are constructed without going through the cocycle / coboundary intermediate stage — i.e. the modular-form analog of writing down on-shell amplitudes directly via BCFW or unitarity rather than gauge-fixing an off-shell Lagrangian — would seem to require an independent characterisation of the cuspidal-cohomology coordinates of $H_{\text{cusp}}^1(\Gamma; V_n)$. We flag this as an open question; what the modular-form analog of the on-shell S-matrix programme might be is, in our view, a worthy direction.

References

- [1] F. Brown, *A class of non-holomorphic modular forms III*, arXiv:1710.07912 (2017).
- [2] N. Diamantis and L. Rolin, *Period polynomials, derivatives of L-functions, and zeros of polynomials*, arXiv:1707.04814 (2017).
- [3] K. Bringmann, K.-H. Fricke, and Z. Kent, *Special L-values and periods of weakly holomorphic modular forms*, Ramanujan J. **24** (2011), 119–139.
- [4] D. A. McGady, *L-functions for meromorphic modular forms*, arxiv:1806.abcde
- [5] W. Kohnen and D. Zagier, *Modular forms with rational periods*, in *Modular Forms* (R.A. Rankin, ed.), Ellis Horwood, 1984, pp. 197–249.
- [6] J. Lewis and D. Zagier, *Period functions for Maass wave forms. I*, Annals of Mathematics, 153 (2001), 191–258.
- [7] Y.-t. Huang and D. McGady, *Consistency Conditions for Gauge Theory S Matrices from Requirements of Generalized Unitarity*, Phys. Rev. Lett. **112** (2014) 241601, arXiv:1307.4065 [hep-th].
- [8] W.-M. Chen, Y.-t. Huang, and D. A. McGady, *Anomalies without an action*, arXiv:1402.7062 [hep-th] (2014).
- [9] `period_extraction_DR_general.ipynb` (executed Jupyter notebook). Companion source file; implements `extract_periods(coeffs_dict, tau0, k)`: the single q -series-agnostic function reproducing the $\omega^\pm, \eta^\pm$ values of Corollaries 1.3, 1.4 and the basepoint-independence check of Corollary 6.2, at 45-digit `mpmath` precision.
- [10] `verify_bernoulli_formula.py`, companion script. Confirms agreement of the closed-form Bernoulli operator (10) with the upper-triangular recursion of Proposition 3.1 on every p_m , $m = 0, \dots, 9$, to relative residual $\sim 10^{-42}$ at 45-digit working precision.
- [11] `verify_omega_plus_formula.py`, companion script. Evaluates each of the nine equivalent forms of $\omega^\pm(\Delta)$ in (16) by computing $L(\Delta, s)$ via the BFK incomplete-gamma sum, confirming agreement with Brown’s reported values to relative error $< 6 \times 10^{-42}$. Source of the $L_{\text{reg}}(\Delta, s)$ table in §5.
- [12] `compute_period_kernels.py`, companion script. Computes the nine universal $\mathbb{Q}$ -polynomial kernel pairs (k_S^j, k_T^j) of Table 1 exactly via sympy rationals, by applying the δ_T^{-1} Bernoulli operator monomial-by-monomial as in (15).

- [13] `compute_numerical_polynomials.py`, companion script. Computes the explicit modified cocycle values $C_S'^f$ of Remark 6.3 at $\tau_0 = 0.3 + 1.2i$ for $f \in \{\Delta, \widehat{\Delta}\}$ at 30-digit `mpmath` precision. Verifies period extraction from all nine interior monomials to lockstep agreement.
- [14] `dr_b_Lfunction_complex_s.py`, companion script. Computes $\Lambda^*(\Delta, s)$ both via the BFK incomplete-gamma reference sum and via the Hurwitz-zeta DR-B finite-contour functional of Theorem 8.1. Verifies (i) the gap identity (22) for $s = 1, \dots, 11$ at relative residual $\sim 10^{-54}$; (ii) the integer- s reduction of Theorem 8.1 at all $s \in \{1, \dots, 11\}$ *including* the two boundary values $s = 1, 11$ unreachable by the bounded-degree V_{10} Bernoulli inversion; (iii) the non-integer test points $s \in \{-0.3, 0.5, 1.5, 2.5, 3.5, 4.5, 5.5, 6, 6.5\}$ and the complex test points $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$; and (iv) basepoint independence at fixed $s = 3 + 1.5i$ across $\tau_0 \in \{0.3 + 1.2i, 1.5i, -0.4 + 0.8i, 0.2 + 2i\}$. All checks agree at relative residual $\sim 10^{-53}$ at 50-digit `mpmath` precision.
- [15] NIST Digital Library of Mathematical Functions, F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds., <https://dlmf.nist.gov/>, Release 1.2.x. Chapters used: §8 (incomplete gamma functions) and §25.11 (Hurwitz zeta function).
- [16] `verify_s6.py`, companion script. Per-mode verification at $\tau_0 = i$ of the equality between the DR-B integrand (27) and the BFK incomplete-gamma summand at $s = 6$ for $n \in \{-1, 1, 2, 3, 4, 5\}$, at 50-digit `mpmath` precision.
- [17] `verify_multi_s.py`, companion script. Per-mode verification of (27) versus the BFK summand at $s \in \{1, 3, 6, 9, 11\}$ — including the boundary integers $s = 1, 11$ unreachable by the bounded-degree V_{10} Bernoulli inversion — for $n \in \{-1, 1, 2, 3\}$, at 50-digit `mpmath` precision.
- [18] Long-form companion document `analytic_periods_weight12.tex` / `.pdf`. Same mathematical content as the present note with extensive pedagogical discussion of cocycles, coboundaries, the Bernoulli-operator derivation, and worked numerical polynomials. The present note is the math-paper-style restatement with (2) as the front matter.